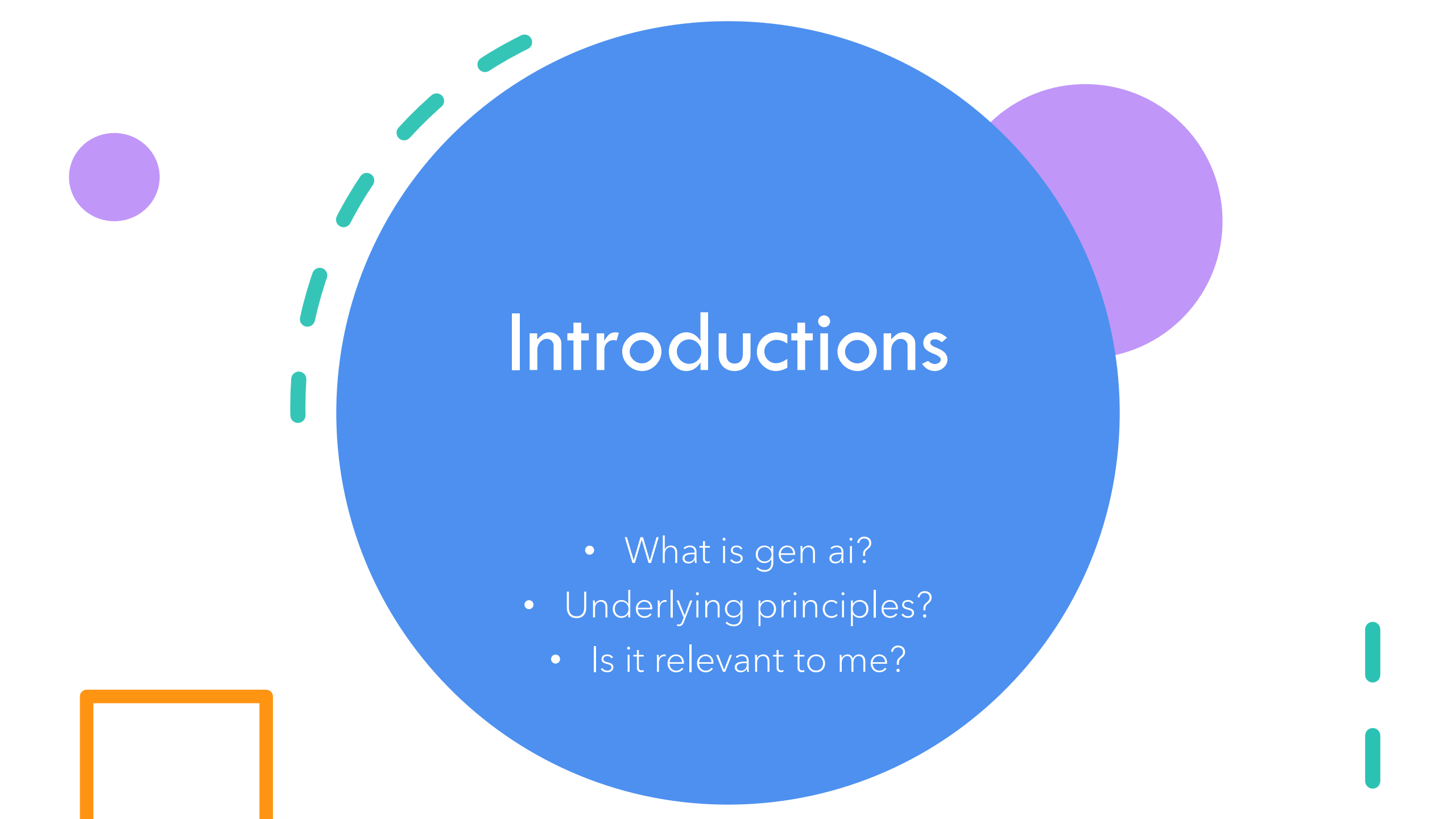




Introduction to Generative AI

Simplifying Software Development

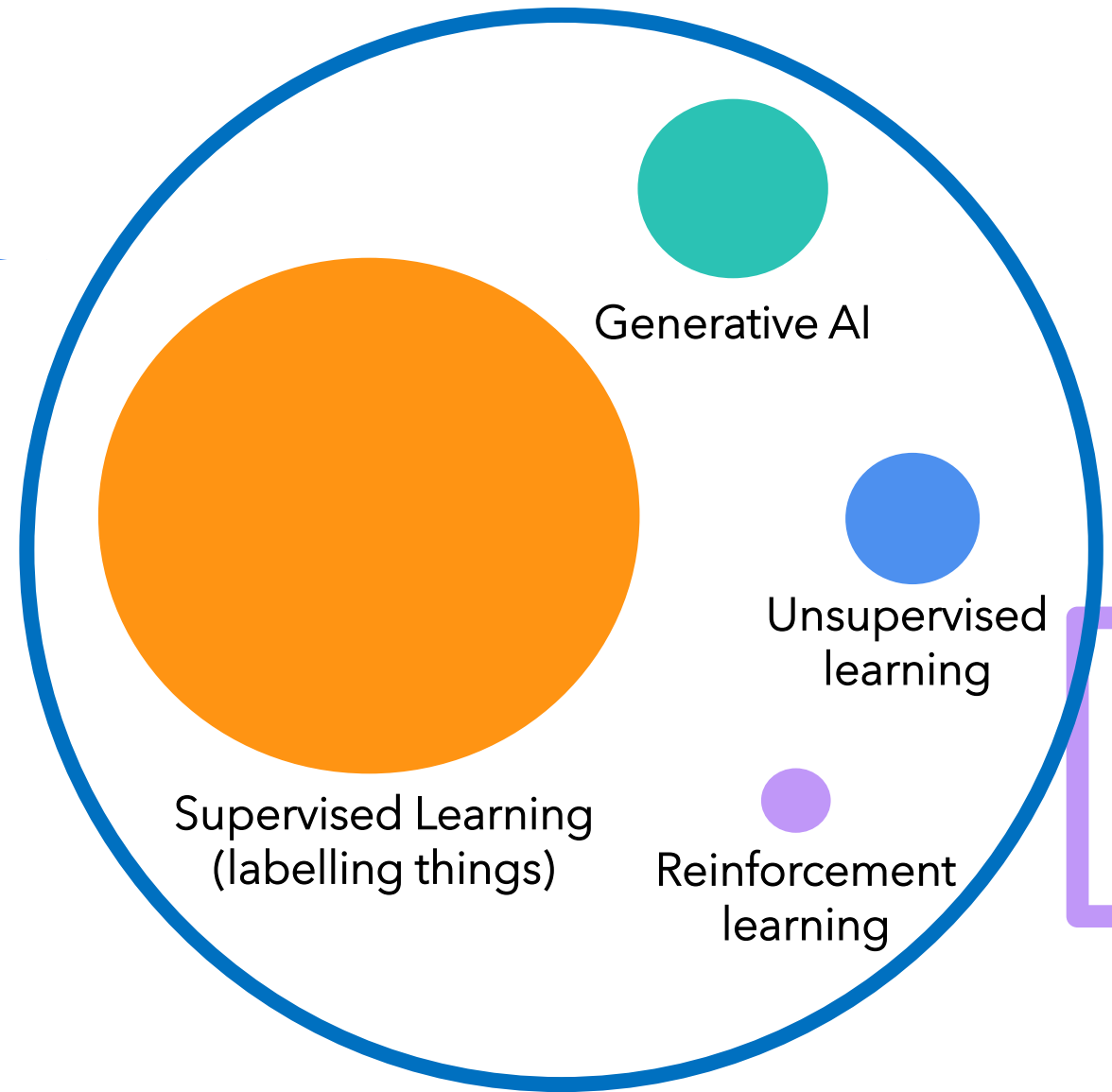


Introductions

- What is gen ai?
- Underlying principles?
- Is it relevant to me?

AI is a set of tools

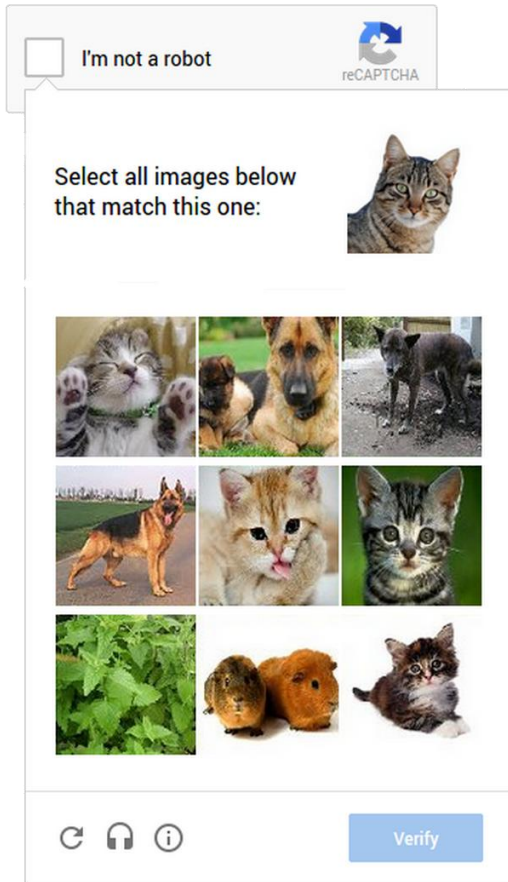
Artificial Intelligence



Supervised Learning

Supervised learning is a machine learning technique that uses labeled data sets to train artificial intelligence (AI) models to identify the underlying patterns and relationships. The goal of the learning process is to create a model that can predict correct outputs on new real-world data.

Real time Example



- Data scientist prepare labelled data sets containing numerous animals and their categorization like dogs, cats, etc.
- Model's algorithm will then attempt to identify based on provided input to receive designated output.
- The results are then measured against the actual values in data sets for verification.
- If not, the training cycle continues until the model's performance has reached a satisfactory level of accuracy.
- The principle of generalization refers to a trained model's ability to make appropriate predictions on new data from the same distribution as its training data.

Text generation with supervised Learning

LLMs don't "know" facts — they predict the next best word based on patterns learned from data.

Prompt:

I love eating _____.

AI output.

Dosa/Idli/Fruits

LLMs are built by using supervised learning ($A \rightarrow B$) to repeatedly predict the next word.

My favorite food is dosa at CTR.

Input(A)

AI Output(B)

My favorite food is

dosa

My favorite food is dosa

at

My favorite food is dosa at

CTR



When you train a very large AI system on a lot of data(hundreds of billions of words), we get a large Language Model like ChatGPT.

Generative AI

*Artificial intelligent systems that can produce high quality content, specifically **texts, images, and audio**. Examples:*

❖ ChatGPT/OpenAI

❖ BARD/Google

❖ Bing Chat/Microsoft

What prompting an LLM can do?

Can a fresh college graduate follow the instructions in the prompt to complete the task?

For clarity, assume:

- No access to the internet or other resources.
- No training specific to your company/business.
- No memory of previous tasks completed
 - You get a fresh college grad each time!

Limitations

- Knowledge cutoffs.
- Hallucinations : Making things up.
 - Hallucinations have had serious consequences.
- The input (and output) length is limited.
 - Total amount of text you can give is limited.
 - An LLM's context length is the limit on the total input+output size.
- Generative AI does not work well with structured (tabular) data, prefer supervised learning.
- Gen AI works best with unstructured data referring to text, images, audio, videos, etc.
- Bias and Toxicity

Limitations

Example: Gender Bias in LLM Outputs

Prompt:

"The surgeon walked to the parking lot and took out ____."

Biased LLM Output (Common Pattern):

"his car keys."

Why This Is Biased:

- Assumes **surgeon = male**
- Reinforces historical and societal stereotypes
- Gender was **never specified** in the prompt

Prompt:

"The nurse walked to the parking lot and took out ____."

Biased LLM Output (Common Pattern):

"her phone."

Why This Is Biased:

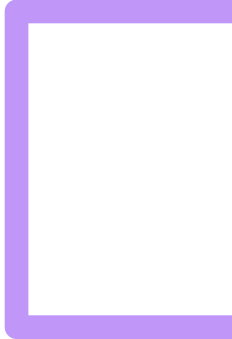
- Assumes **nurse = female**
- Reflects skewed training data, not reality

Why LLMs Make These Assumptions

- Trained on **real-world text**, which contains biases
- Media and historical data over-represent:
 - Male surgeons
 - Female nurses
- Model learns **patterns**, not fairness



Tips for prompting

- ❖ Be detailed and specific
 - ❖ Guide the model to think through its
answer
 - ❖ Experiment and iterate
- 
- 

Tips for prompting

Give sufficient context for LLM to complete the task

✗ Less Context-Based Prompt

"Write an email to a client about a delay."

✓ Detailed & Specific Prompt

"Write a professional email to an enterprise client explaining a 2-day project delay caused by a vendor issue. Apologize briefly, reassure delivery by Friday, and keep the tone calm and confident."

Tips for prompting

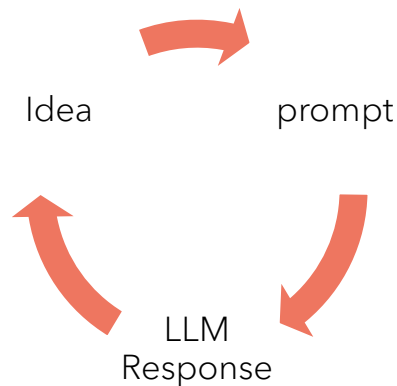
Guide the model to think through its answer "Chain of thought"

✗ Prompt Without Reasoning Guidance
"Should our company adopt Generative AI?"

✓ Prompt That Guides Thinking (Step-by-Step)
"Should our company adopt Generative AI?
Think through the answer step by step by:
1. Identifying key business benefits
2. Highlighting potential risks
3. Suggesting 2 practical use cases
4. Concluding with a recommendation for leadership"

Tips for prompting

Iterate improving your prompt



Prompting process

- Be clear and specific
- Think about why results isn't giving the desired output
- Refine your prompt
- Repeat

Tip: Don't overthink the initial prompt. Just try something quickly and improve it over time!

Caveats:

(i) confidential information

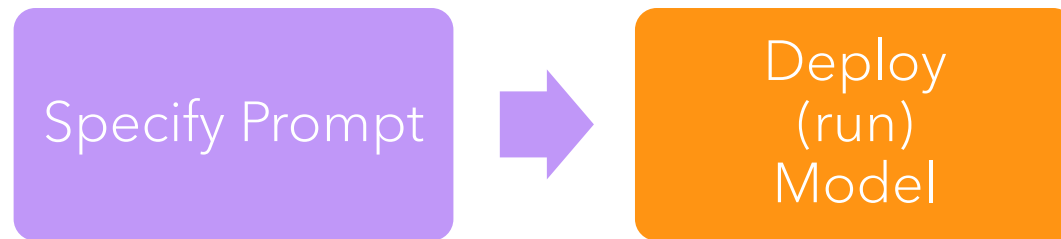
(ii) Whether you trust the LLM output, (hallucination)

Using Generative AI in software applications

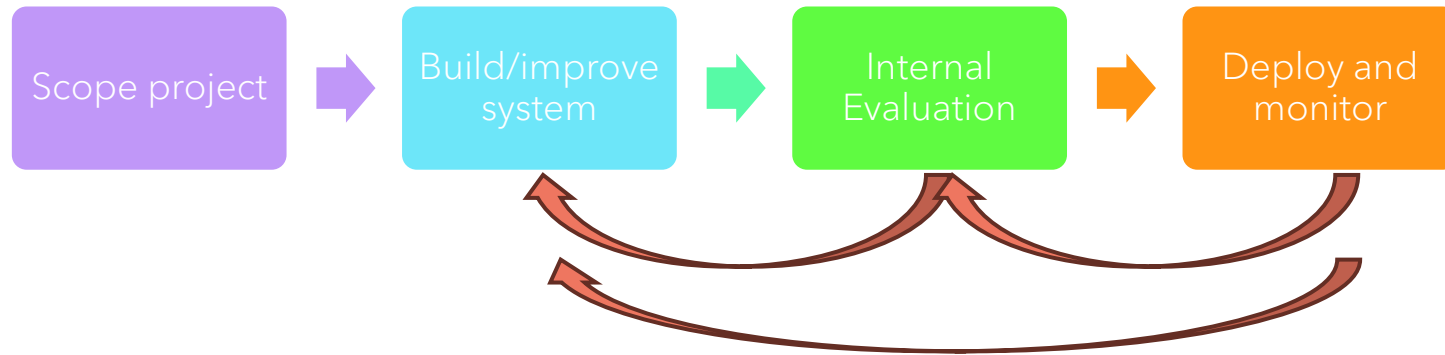
Supervised learning → approx.(1+3+3 months)



Prompt-based AI → approx.(days/weeks)



Life cycle of a gen AI project



Building gen AI is a highly empirical (experimental) process where we repeatedly find and fix mistakes.

Cost intuition:

How much does it cost?

Provider / Model	Approx. Cost per 1,000 Input Tokens	Approx. Cost per 1,000 Output Tokens
OpenAI (GPT-5 family)	~\$0.00125	~\$0.01
Google Gemini 2.5 Flash-Lite (PaLM-based)	~\$0.00010	~\$0.00040
Amazon Titan Text Lite (via AWS Bedrock)	~\$0.00030	~\$0.00040

Retrieval Augmented Generation

Combines retrieval and generation to make LLMs accurate, factual and grounded in real data.

LLMs alone can:

- “hallucinate” facts
- miss company-specific knowledge
- provide outdated info

RAG fixes this by using real documents as reference.

Exercise:

1. Ask any LLM to answer a specific question to a document, only you have access to.
2. Use RAG by providing the relevant document and then see a difference in the output.

Pre-training

Pre-training = Teaching the model language basics

- The model learns **grammar, facts, reasoning patterns, and general knowledge**
- Done using **massive datasets** (books, articles, websites, code)

How it works:

- Model is given text
- Task: **predict the next word**
- It learns patterns like:
 - grammar
 - meaning
 - context

Example:

- Like teaching a student *general literacy* before they specialize.

Pre-training

- Costly affair
- High economy
- Open-source availability

Fine-tuning

Fine-tuning = Customizing the model for a specific job

- Takes the pre-trained model and trains it further using **specific data**
- Makes the model better for:
 - customer support
 - legal drafting
 - medical summaries
 - internal knowledge

Example:

- Like training a generalist employee to become a specialist.

Why fine tune?

- To carry out a task that isn't easy to define in a prompt.
 - Example: Summarize in a certain style/ Reporting.
 - Example: Mimicking a writing or speaking style.
- To help LLM gain specific knowledge.
 - DOA, DOD, pt, etc. for a medical record.
- To get smaller model to perform a task.
 - 100B+ model parameters VS 1B parameters.

Model Size

Model Size	Best For	Pros	Cons
Small / Compact	Mobile apps, chatbots, real-time tools	Fast, cheap, low latency	Less accurate, limited reasoning
Medium	Internal tools, customer support	Balanced cost & quality	May struggle with complex tasks
Large / High-capacity	Research, complex analysis, coding	High accuracy, strong reasoning	Expensive, slower, higher compute
Extra Large / Cutting-edge	High-stakes decisions, advanced automation	Best performance	Very expensive, high latency

Open or closed source?

Aspect	Open Source	Closed Source
Access to Code	Yes	No
Customization	High	Limited
Cost	Often lower	Usually higher
Speed of Innovation	Fast (community-driven)	Controlled (vendor-driven)
Support & Reliability	Community-based	Professional support
Data Privacy	Can be self-hosted	Depends on vendor
Security Control	High (self-managed)	Vendor-managed
Best For	Internal tools, research, privacy	Enterprise apps, compliance, scalability

Instruction tuning

Instruction tuning is the process of training a model to follow **human-style** instructions better.

Instead of just predicting the next word, the model learns:

- how to respond to prompts
- how to follow formatting rules
- how to be helpful and safe

Reinforcement learning from Human feedback ()

Helpful, Honest, Harmless

Step 1. Train on answer quality (reward) model

Prompt:

"Advise me on how to apply for a job."

Response	Score(Reward)
I'm happy to help! Here are some steps to follow [...]	5
Just try your best!	3
It's hopeless -- why bother?	1

Step 2: Have LLM generate a lot of answers. Further train it to generate more responses that get high scores.

Tool use for food order taking

Send me a burger

Passed to restaurant ordering system:

Item: Burger
UserID: 9876
Address: 1234 My Street

ORDER(Burger, 9876,
1234 My street)

User message:

Ok, It's on its way!

Displayed to user as response

Ok, It's on its way!

Agents

- Use LLM to choose and carry out complex sequence of actions
- Cutting edge area of AI research
- Example

Task: "Book a flight for next week"

Agent:

1. Searches flights
2. Compares prices
3. Books ticket
4. Sends confirmation

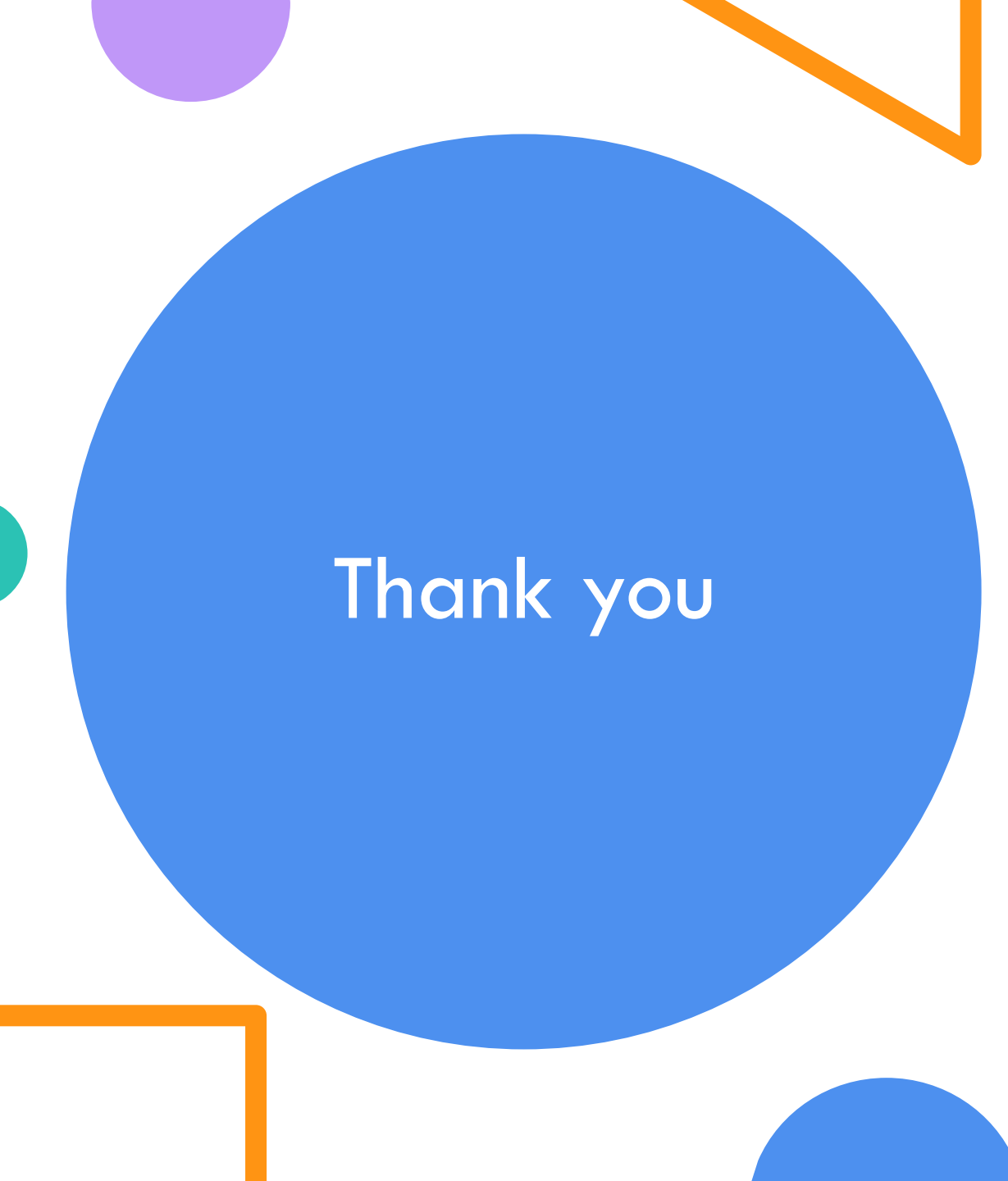
SEARCH("Flight provider")

VISIT(www.booking.com)

VISIT(www.makemy trip.com)

• • •

Thanks, here's your flight details [...]



Thank you

Anuj Mundhra

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